Robustness Ablation of Deep Vision Models under Clean and Corrupted Validation

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***Abstract*—This report documents the full experimental pipeline implemented for Assignment–02 on robustness-aware model selection for image classification. Four architectures — VGG16-BN, ResNet50, ConvNeXt-Tiny, and ViT-B/16 — were trained on CIFAR-10, Fashion-MNIST, and ImageNet-100. A fixed 20% validation split was used throughout, and model selection was studied under both clean validation and corrupted validation. The implemented accelerated protocol used five corruption types (Gaussian noise, motion blur, fog, brightness, and JPEG compression) at severity level 2, with selector policies `corr_k1_s2`, `corr_k3_s2`, and `corr_k5_s2`. Results show that corruption-aware validation usually improves corrupted test performance with very small changes in clean accuracy. Across all 12 model–dataset pairs, the best corruption-aware selector improved mean corrupted test accuracy in 11 out of 12 cases. The strongest clean and robust performance in the available runs was consistently achieved by the VGG model on all three datasets. A separate intermediate feature-perturbation study was also executed for ResNet across the three datasets.**

***Index Terms*—robustness, corrupted validation, image classification, VGG, ResNet, ConvNeXt, Vision Transformer, CIFAR-10, Fashion-MNIST, ImageNet-100**

I. Assignment Objective and Scope

This is a systematic robustness study of multiple deep vision architectures under both clean and corrupted validation settings. The required work can be broken into the following questions.

1) Train VGG, ResNet, ConvNeXt, and ViT on CIFAR-10, Fashion-MNIST, and ImageNet-100.
2) Create a fixed 20% validation split from the training set and keep it constant across experiments.
3) Compare checkpoint selection under clean validation and corrupted validation.
4) Study how much corruption should be used in the validation procedure.
5) Report classification accuracy on clean and corrupted test sets.
6) Provide visual analysis of the effect of the validation setting.
7) Explore intermediate feature perturbation as an additional robustness analysis.

This report follows that structure directly. It explains what was implemented, how the experiments were organized, and what was observed from the generated result tables. No numbers are invented; every reported result in this document is taken from the uploaded CSV files produced by the experiment pipeline.

II. Experimental Setup

A. Hardware and Software Context

The executed notebooks reported the following environment details during the reporting stage:

- PyTorch version: 2.6.0+cu124
- CUDA available: True
- GPU: NVIDIA H100 NVL

B. Datasets

Three datasets were used:

- `cifar10`
- `fashion_mnist`
- `imagenet100`

The local preparation stage produced a `train/` and `val/` folder structure for ImageNet-100. CIFAR-10 and Fashion-MNIST were prepared locally through `torchvision`-based dataset loaders. Fashion-MNIST was converted to RGB so that all four backbones could share a common three-channel input interface.

C. Models

The following backbones were used:

- `vgg`: VGG16-BN
- `resnet`: ResNet50
- `convnext`: ConvNeXt-Tiny
- `vit`: ViT-B/16

D. Input Preprocessing

All datasets were standardized to an input size of 224×224 pixels. The main motivation was to keep the input pipeline consistent across all backbones, especially because ViT-B/16 and the ImageNet-oriented CNNs are naturally configured for ImageNet-scale image sizes. Fashion-MNIST images were first converted to RGB and then resized.

E. Validation Split

A fixed and stratified 20% validation split was created from the training set for each dataset. The split indices were stored as JSON files:

- `splits/cifar10_seed42_val20.json`
- `splits/fashion_mnist_seed42_val20.json`
- `splits/imagenet100_seed42_val20.json`

This is important because the assignment explicitly requires the validation set to remain fixed once created.

F. Accelerated Corruption Protocol

The final implementation used an accelerated five-corruption protocol rather than a larger 15-corruption sweep. This decision was taken for computational feasibility after the base pipeline had already been developed and debugged.

The active corruption set was:

1) `gaussian_noise`
2) `motion_blur`
3) `fog`
4) `brightness`
5) `jpeg_compression`

All corrupted validation and corrupted test evaluations used severity 2. The compared selectors were:

- `clean`
- `corr_k1_s2`
- `corr_k3_s2`
- `corr_k5_s2`

III. Methodology: Step-by-Step Implementation

A. One Training Trajectory, Multiple Selectors

For each dataset–model pair, the model was trained once. During training, validation was evaluated under several selector policies:

- clean validation
- corrupted validation averaged over the first 1 corruption
- corrupted validation averaged over the first 3 corruptions
- corrupted validation averaged over all 5 selected corruptions

This design is computationally more efficient than retraining separate models for each validation policy, while still answering the assignment's central question: does the best checkpoint change when the validation criterion changes?

B. Per-Epoch Procedure

At each epoch, the implemented training loop followed the same sequence:

1) run one full training epoch,
2) evaluate on the clean validation split,
3) evaluate on the corrupted validation split for each active corruption,
4) compute selector scores for `corr_k1_s2`, `corr_k3_s2`, and `corr_k5_s2`,
5) update the best checkpoint for each selector independently,
6) log the epoch summary to the history table.

C. Checkpoint Evaluation

After training completed for a pair, each saved best-selector checkpoint was reloaded and evaluated on:

- the clean test set,
- the corrupted test set under the same five corruption types,
- the mean corrupted test accuracy.

The main per-selector metrics stored in the result tables were:

- **best_val_score**
- **test_clean_acc**
- **test_corrupt_mean_acc**
- **robustness_gap** = clean accuracy − mean corrupted accuracy

D. Visual Analysis

A dedicated notebook was used to generate visual analysis for all selectors of all completed pairs. For each selector, the following files were created:

- clean confusion matrix
- corrupted confusion matrix
- clean feature-space projection
- corrupted feature-space projection
- clean feature maps
- corrupted feature maps
- clean classification report CSV
- corrupted classification report CSV

The reporting log showed successful generation for all 12 pairs and all 4 selector settings, i.e., 48 selector-specific visualization folders in total.

E. Intermediate Feature Perturbation

A separate notebook executed an additional study in which Gaussian noise was injected into internal representations of ResNet. Three locations were evaluated:

- early layers,
- middle layers,
- late layers.

This part was run for CIFAR-10, Fashion-MNIST, and ImageNet-100 using the ResNet model.

IV. Question-by-Question Answers with Results

A. Q1. Training All Required Models on All Required Datasets

The experiment grid contained 12 pairs:

$$4 \text{ models} \times 3 \text{ datasets} = 12 \text{ runs}.$$

The final aggregated CSV files confirm that all 12 pairs produced report rows for both clean and corruption-aware selection.

B. Q2. Fixed 20% Validation Split

The fixed split requirement was satisfied through saved JSON index files. Since all selector policies reuse the same train/validation partition, model comparison remains fair.

C. Q3. Clean Validation versus Corrupted Validation

Table I reports the clean-selector results. Table II reports the best corruption-aware selector for each pair.

A direct comparison between clean-selected and best-corruption-selected checkpoints is given in Table III.

The results show that corruption-aware selection improved mean corrupted test accuracy in 11 out of 12 pairs. The only exception was `fashion_mnist` with `resnet`, where the change was slightly negative (-0.06 percentage points), which is practically negligible compared with the much larger positive gains seen in several other settings.

TABLE I

RESULTS OBTAINED WHEN THE BEST CHECKPOINT IS SELECTED USING CLEAN VALIDATION. ALL ACCURACIES ARE PERCENTAGES.

Dataset	Model	Best Epoch	Best Val.	Clean Test	Mean Corrupt Test	Gap
cifar10	convnext	24	84.24	83.56	73.21	10.35
cifar10	resnet	25	66.34	66.11	44.21	21.90
cifar10	vgg	23	89.71	88.73	76.45	12.28
cifar10	vit	25	72.37	72.21	60.75	11.46
fashion_mnist	convnext	25	93.72	92.91	66.70	26.21
fashion_mnist	resnet	3	69.57	69.59	39.76	29.83
fashion_mnist	vgg	24	94.98	94.29	77.15	17.14
fashion_mnist	vit	25	90.28	90.01	67.07	22.94
imagenet100	convnext	15	54.85	55.15	34.69	20.46
imagenet100	resnet	20	42.23	42.07	18.74	23.33
imagenet100	vgg	20	68.44	69.08	36.98	32.10
imagenet100	vit	14	47.16	47.03	32.06	14.97

TABLE II

BEST CORRUPTION-AWARE SELECTOR FOR EACH DATASET–MODEL PAIR. THE SELECTOR SHOWN IS THE CORRUPTION POLICY THAT PRODUCED THE HIGHEST CORRUPTED TEST MEAN AMONG `corr_k1_s2`, `corr_k3_s2`, AND `corr_k5_s2`. ALL ACCURACIES ARE PERCENTAGES.

Dataset	Model	Selector	Best Epoch	Best Val.	Clean Test	Mean Corrupt Test	Gap
cifar10	convnext	corr_k3_s2	18	68.46	82.94	73.40	9.54
cifar10	resnet	corr_k3_s2	24	34.77	66.16	44.54	21.62
cifar10	vgg	corr_k1_s2	20	76.78	88.28	76.60	11.68
cifar10	vit	corr_k3_s2	25	56.68	72.21	60.84	11.37
fashion_mnist	convnext	corr_k3_s2	19	71.66	92.63	68.76	23.87
fashion_mnist	resnet	corr_k5_s2	3	40.21	69.59	39.71	29.88
fashion_mnist	vgg	corr_k3_s2	18	78.33	94.05	78.41	15.64
fashion_mnist	vit	corr_k5_s2	18	67.62	88.46	67.25	21.21
imagenet100	convnext	corr_k5_s2	13	35.08	54.64	35.39	19.25
imagenet100	resnet	corr_k5_s2	20	19.06	42.07	18.74	23.32
imagenet100	vgg	corr_k5_s2	20	37.21	69.08	37.00	32.08
imagenet100	vit	corr_k5_s2	16	32.49	47.02	32.61	14.41

TABLE III

COMPARISON BETWEEN CLEAN-SELECTED AND BEST-CORRUPTION-SELECTED CHECKPOINTS. THE LAST COLUMN REPORTS THE CHANGE IN MEAN CORRUPTED TEST ACCURACY IN PERCENTAGE POINTS.

Dataset	Model	Best Corrupt Selector	$\text{Clean}_{\text{clean-sel}}$	$\text{Clean}_{\text{corr-sel}}$	$\text{CorrMean}_{\text{clean-sel}}$	$\text{CorrMean}_{\text{corr-sel}}$	Δ CorrMean
cifar10	convnext	corr_k3_s2	83.56	82.94	73.21	73.40	0.19
cifar10	resnet	corr_k3_s2	66.11	66.16	44.21	44.54	0.33
cifar10	vgg	corr_k1_s2	88.73	88.28	76.45	76.60	0.15
cifar10	vit	corr_k3_s2	72.21	72.21	60.75	60.84	0.09
fashion_mnist	convnext	corr_k3_s2	92.91	92.63	66.70	68.76	2.06
fashion_mnist	resnet	corr_k5_s2	69.59	69.59	39.76	39.71	-0.06
fashion_mnist	vgg	corr_k3_s2	94.29	94.05	77.15	78.41	1.26
fashion_mnist	vit	corr_k5_s2	90.01	88.46	67.07	67.25	0.18
imagenet100	convnext	corr_k5_s2	55.15	54.64	34.69	35.39	0.70
imagenet100	resnet	corr_k5_s2	42.07	42.07	18.74	18.74	0.01
imagenet100	vgg	corr_k5_s2	69.08	69.08	36.98	37.00	0.02
imagenet100	vit	corr_k5_s2	47.03	47.02	32.06	32.61	0.55

D. Q4. How Much Corruption Should Be Used in Validation?

Table IV reports mean corrupted test accuracy for all selectors. This table answers the corruption-amount question directly for the implemented accelerated protocol.

Three patterns are clear from Table IV.

First, the best corruption-aware policy is not the same for every pair. For CIFAR-10, the best robust selectors were mostly `corr_k3_s2`, except for VGG where `corr_k1_s2` was strongest. For Fashion-MNIST, VGG and ConvNeXt preferred `corr_k3_s2`, whereas ViT and ResNet preferred `corr_k5_s2`. For ImageNet-100, the best robust selector was consistently `corr_k5_s2`.

Second, using more corruption types in validation generally helped on the harder ImageNet-100 setting.

Third, the improvements in corrupted mean accuracy usually came with a very small reduction in clean test accuracy, which suggests that corruption-aware validation is a reasonable selection strategy if robustness is the priority.

E. Q5. Clean and Corrupted Test Accuracy

The highest clean and robust performance observed in the provided CSV files is summarized below:

- On `cifar10`, the best clean accuracy was obtained by `vgg` (88.73%), while the best mean corrupted accuracy

TABLE IV
MEAN CORRUPTED TEST ACCURACY (IN %) FOR ALL SELECTORS. LARGER IS BETTER.

Dataset	Model	clean	corr_k1_s2	corr_k3_s2	corr_k5_s2
cifar10	convnext	73.21	73.34	73.40	73.14
cifar10	resnet	44.21	32.19	44.54	44.51
cifar10	vgg	76.45	76.60	76.50	76.46
cifar10	vit	60.75	60.64	60.84	60.75
fashion_mnist	convnext	66.70	65.77	68.76	68.67
fashion_mnist	resnet	39.76	34.16	34.11	39.71
fashion_mnist	vgg	77.15	78.30	78.41	78.27
fashion_mnist	vit	67.07	67.06	67.03	67.25
imagenet100	convnext	34.69	30.21	35.25	35.39
imagenet100	resnet	18.74	11.23	18.59	18.74
imagenet100	vgg	36.98	26.81	36.94	37.00
imagenet100	vit	32.06	32.04	32.58	32.61

was also obtained by `vgg` using `corr_k1_s2` (76.60%).

- On `fashion_mnist`, the best clean accuracy was obtained by `vgg` (94.29%), and the best mean corrupted accuracy was again obtained by `vgg` using `corr_k3_s2` (78.41%).
- On `imagenet100`, the best clean accuracy was obtained by `vgg` (69.08%), and the best mean corrupted accuracy was also obtained by `vgg` using `corr_k5_s2` (37.00%).

Thus, among the completed runs, VGG was the strongest model on all three datasets in both clean and corruption-aware evaluation.

F. Q6. Visual Analysis

The visual-analysis notebook finished successfully for all selectors of all 12 pairs. Figure placeholders are provided below so that the final Overleaf project can include the exported images.

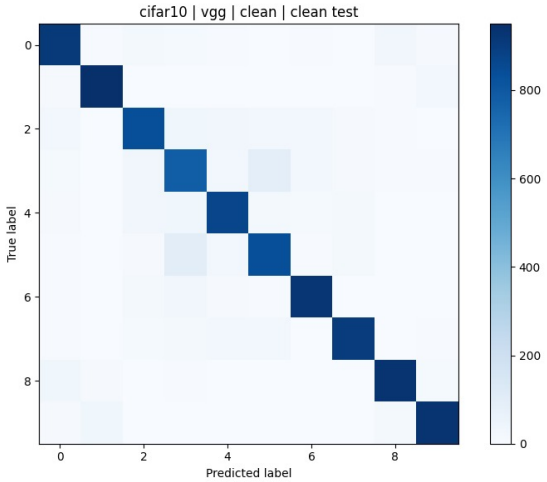

Fig. 1. Suggested figure: clean confusion matrix for the CIFAR-10 VGG clean-selected checkpoint.

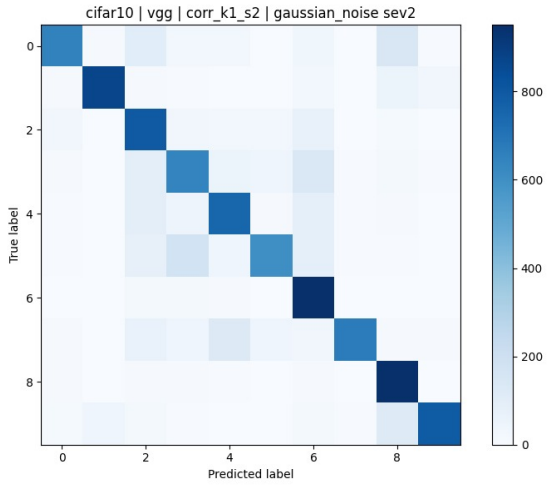

Fig. 2. Suggested figure: feature-space projection for the CIFAR-10 VGG corruption-aware checkpoint (`corr_k1_s2`) under Gaussian-noise corruption.

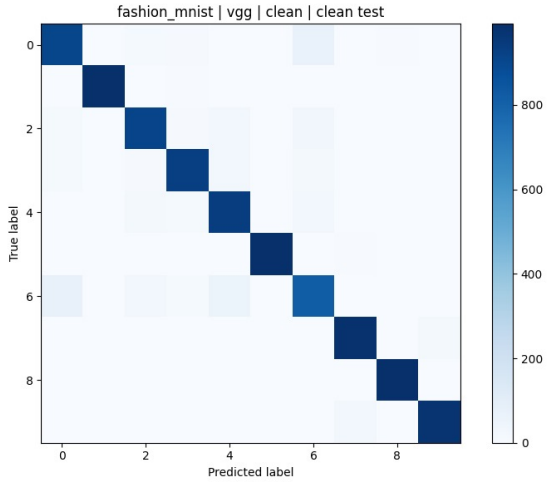

Fig. 3. Fashion-MNIST VGG clean confusion matrix.

Table V summarizes the feature-perturbation ablation that was run for ResNet.

TABLE V
INTERMEDIATE FEATURE-PERTURBATION RESULTS FOR RESNET. ALL VALUES ARE PERCENTAGES.

Dataset	Where	Clean Test	Mean Corrupt Test	Gap
cifar10	early	47.89	35.75	12.14
cifar10	late	47.00	35.58	11.42
cifar10	mid	48.08	35.48	12.60
fashion_mnist	early	55.72	38.34	17.38
fashion_mnist	late	77.64	44.29	33.35
fashion_mnist	mid	77.52	44.33	33.19
imagenet100	early	19.83	11.50	8.34
imagenet100	late	19.91	11.43	8.48
imagenet100	mid	19.90	11.47	8.43

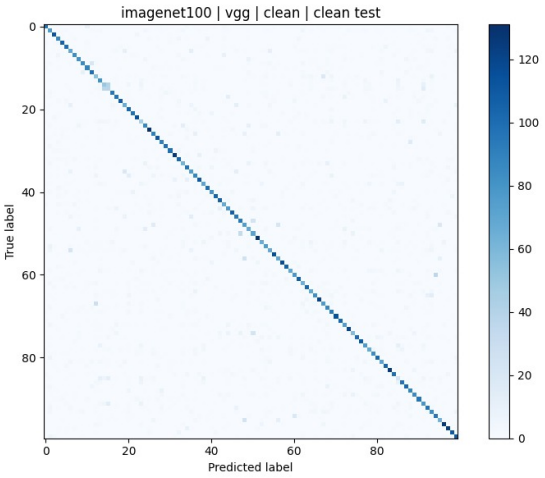

Fig. 4. Suggested figure: clean feature-space projection for the ImageNet-100 VGG clean-selected checkpoint.

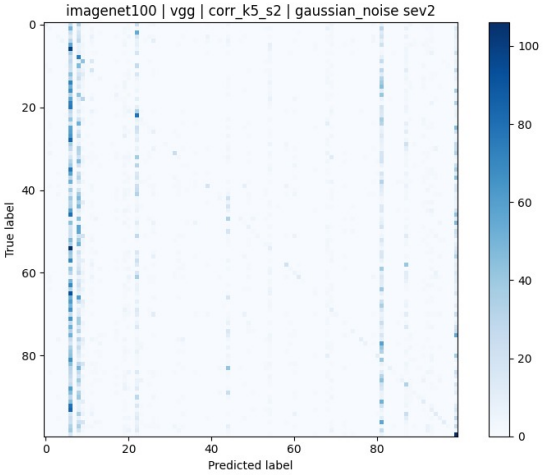

Fig. 5. Suggested figure: corrupted confusion matrix for the ImageNet-100 VGG `corr_k5_s2` checkpoint.

G. Q7. Intermediate Feature Perturbation

The feature-perturbation study should be interpreted carefully because it was run only for ResNet. Even so, it still provides useful diagnostic information. For CIFAR-10, the early, middle, and late perturbations all produced similar corrupted mean accuracies around 35.5%–35.8%. For Fashion-MNIST, late and mid perturbation produced much larger robustness gaps (above 33 percentage points) than early perturbation (17.38 percentage points). For ImageNet-100, all three perturbation locations produced very similar results, with corrupted mean accuracy close to 11.5%.

V. DISCUSSION

Several conclusions follow from the experimental results.

A. Corruption-Aware Validation Usually Helps

The clearest finding is that the best corruption-aware selector improved mean corrupted test accuracy in 11 of the 12 pairs. The gains were especially meaningful for:

- Fashion-MNIST + ConvNeXt: +2.06 points
- Fashion-MNIST + VGG: +1.26 points
- ImageNet-100 + ConvNeXt: +0.70 points
- ImageNet-100 + ViT: +0.55 points

These are not trivial fluctuations; they show that the validation protocol influences which checkpoint is considered "best".

B. VGG Was Surprisingly Strong

In these runs, VGG achieved the highest clean test accuracy and the highest mean corrupted test accuracy on all three datasets. This does not mean VGG is always the best architecture in general. It only means that under this exact experimental setup — input resolution, optimizer choices, runtime budget, and corruption set — VGG emerged as the strongest performer.

C. ImageNet-100 Was Much Harder

ImageNet-100 produced the lowest corrupted mean accuracies and the largest robustness gaps. Even the strongest ImageNet-100 setting, VGG with `corr_k5_s2`, reached only 37.00% mean corrupted test accuracy, whereas Fashion-MNIST VGG with `corr_k3_s2` reached 78.41%.

D. Effect on Clean Accuracy Was Usually Small

In most cases, moving from clean selection to the best corruption-aware selector caused only a small drop in clean accuracy, and in a few cases there was no drop at all. This is important because it means robustness can often be improved without a severe penalty in clean performance.

VI. Limitations

This report should be read with the following limitations in mind.

- The implemented protocol uses five corruption types rather than a larger 15-corruption benchmark.
- The feature-perturbation study was run only for ResNet.
- The visual-analysis section depends on figure files that are generated externally and must be uploaded to Overleaf separately.
- The results are specific to the executed setup and should not be interpreted as universal rankings of the four backbone families.

VII. Conclusion

This assignment was completed as a full robustness-evaluation pipeline rather than as a single training notebook. The work covered model training, fixed-split validation, clean vs corrupted checkpoint selection, selector-wise evaluation, aggregation of report tables, full selector-wise visual generation, and an additional intermediate feature-perturbation study.

From the actual result files, the most important conclusions are:

1) Clean validation and corruption-aware validation do not always select the same checkpoint.
2) Corruption-aware validation improved mean corrupted test accuracy in 11 out of 12 dataset–model pairs.
3) The best corruption-aware policy depended on the dataset and model, with `corr_k3_s2` and `corr_k5_s2` being the most common winners.
4) In the completed runs, VGG was the strongest model on all three datasets in both clean and robust performance.

Overall, the results support the main assignment claim: if robustness matters, model selection should not rely only on clean validation accuracy.

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